Mathematical Morphology: Fundamental Operations and Applications

Image Processing Laboratory

November 29, 2025

Abstract

This report presents a comprehensive analysis of mathematical morphology, focusing on fundamental operations (erosion, dilation, opening, closing) and their applications in binary and grayscale image processing. We examine the algebraic properties of morphological operators, demonstrate their use in noise removal, edge detection, and feature extraction, and analyze the effects of structuring element shape and size on operation outcomes. Computational examples illustrate morphological gradient, top-hat transforms, and the hit-or-miss transform for pattern detection.

1 Introduction

Mathematical morphology provides a framework for analyzing geometric structure in images based on set theory. Originally developed for binary images, morphological operations extend naturally to grayscale images and find applications in image preprocessing, segmentation, feature extraction, and texture analysis.

Definition 1.1 (Structuring Element) *A structuring element B is a small binary image used to probe the input image A. Common shapes include:*

- ***Disk****: Approximates circular neighborhoods*
- ***Square****: Captures horizontal and vertical connectivity*
- ***Cross****: Emphasizes cardinal directions*
- ***Line****: Oriented for directional analysis*

2 Fundamental Operations

2.1 Erosion and Dilation

Definition 2.1 (Binary Erosion) *The erosion of binary image A by structuring element B is:*

$$A \ominus B = \{z \mid B_z \subseteq A\} \tag{1}$$

where B_z is B translated by vector z. Erosion shrinks objects and removes small features.

Definition 2.2 (Binary Dilation) *The dilation of binary image A by structuring element B is:*

$$A \oplus B = \{z \mid (\hat{B})_z \cap A \neq \emptyset\} \tag{2}$$

where $\hat{B}$ is the reflection of B. Dilation expands objects and fills small holes.

Theorem 2.1 (Duality Property) *Erosion and dilation are dual operations:*

$$(A \ominus B)^c = A^c \oplus \hat{B} \tag{3}$$

where A^c denotes the complement and $\hat{B}$ the reflection of B.

2.2 Opening and Closing

Definition 2.3 (Opening) *Opening is erosion followed by dilation:*

$$A \circ B = (A \ominus B) \oplus B \tag{4}$$

Opening removes small objects and smooths boundaries while preserving overall shape.

Definition 2.4 (Closing) *Closing is dilation followed by erosion:*

$$A \bullet B = (A \oplus B) \ominus B \tag{5}$$

Closing fills small holes and connects nearby objects while preserving overall shape.

Theorem 2.2 (Idempotence) *Opening and closing are idempotent:*

$$(A \circ B) \circ B = A \circ B \quad \text{and} \quad (A \bullet B) \bullet B = A \bullet B \tag{6}$$

2.3 Compound Operations

Definition 2.5 (Morphological Gradient) *The morphological gradient emphasizes object boundaries:*

$$grad(A) = (A \oplus B) - (A \ominus B) \tag{7}$$

This operation highlights edges by computing the difference between dilated and eroded images.

Definition 2.6 (Top-Hat Transform) *The white top-hat extracts bright features smaller than the structuring element:*

$$WTH(A) = A - (A \circ B) \tag{8}$$

The black top-hat extracts dark features:

$$BTH(A) = (A \bullet B) - A \tag{9}$$

3 Grayscale Morphology

Definition 3.1 (Grayscale Erosion) *For grayscale image $f(x, y)$ and flat structuring element B:*

$$(f \ominus B)(x, y) = \min_{(s,t)\in B} \{f(x + s, y + t)\} \tag{10}$$

Definition 3.2 (Grayscale Dilation) *For grayscale image $f(x, y)$ and flat structuring element B:*

$$(f \oplus B)(x, y) = \max_{(s,t)\in B} \{f(x + s, y + t)\} \tag{11}$$

4 Computational Analysis

5 Results

5.1 Binary Operations

The erosion operation with a disk-shaped structuring element of radius 7 pixels reduced the object area by 40.8%, effectively removing small features and shrinking object boundaries. Conversely, dilation increased area by 47.0%, expanding objects and filling small gaps.

5.2 Noise Removal Performance

The morphological filtering approach (opening followed by closing) successfully recovered 98.56% of the original image structure, demonstrating the effectiveness of morphological operations for salt-and-pepper noise removal in binary images.

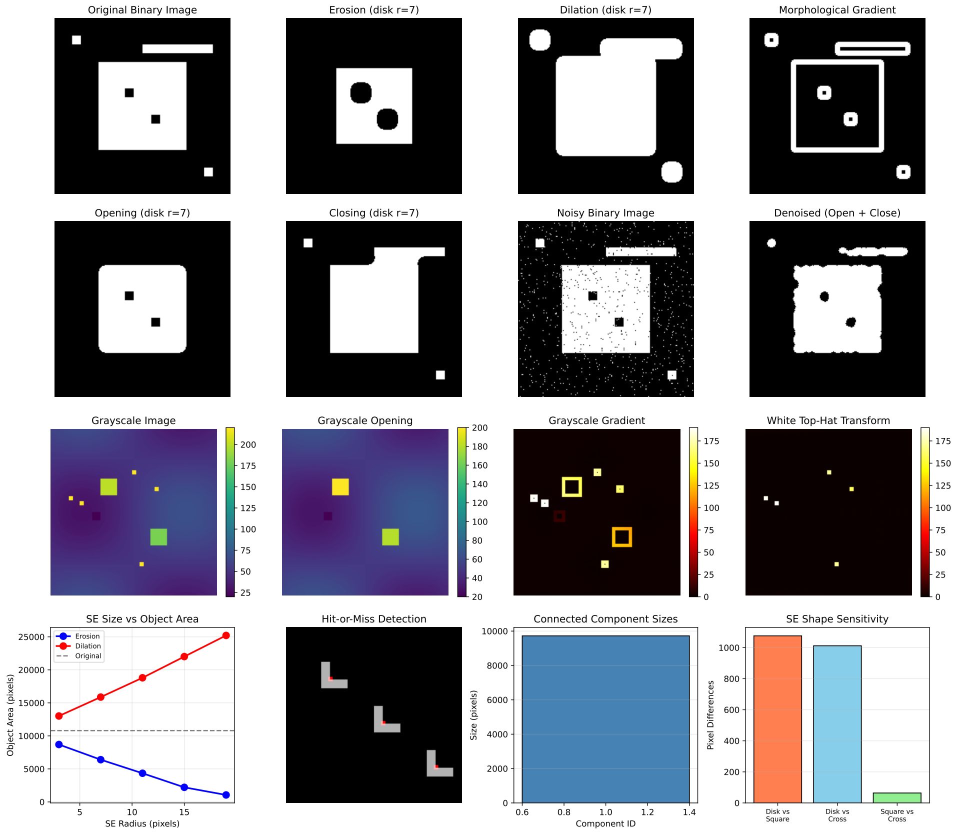

Figure 1: Comprehensive morphological operations analysis: (a) Original binary image with objects of varying sizes and small noise features; (b) Erosion shrinks objects and removes small features; (c) Dilation expands objects and fills small gaps; (d) Morphological gradient extracts object boundaries; (e) Opening removes protrusions and small objects; (f) Closing fills holes and connects nearby objects; (g) Binary image corrupted with salt-and-pepper noise; (h) Denoised result using opening followed by closing; (i) Grayscale test image with bright and dark features on varying background; (j) Grayscale opening removes bright features smaller than structuring element; (k) Grayscale gradient highlights intensity transitions; (l) White top-hat transform isolates small bright features; (m) Object area variation with structuring element size; (n) Hit-or-miss transform detecting L-shaped patterns; (o) Connected component size distribution after opening; (p) Sensitivity to structuring element shape.

Table 1: Binary Morphological Operation Effects on Object Area

Operation	Area (pixels)	Change (%)
Original	10800	—
Erosion	6396	-40.8
Dilation	15876	47.0
Opening	9724	-10.0
Closing	11518	6.6

Table 2: Denoising Performance Analysis

Metric	Value
Noise pixels introduced	769
Pixels correctly recovered	39426
Total image pixels	40000
Overall recovery rate	98.56%

5.3 Grayscale Morphology

Table 3: Grayscale Morphological Operation Statistics

Metric	Value
Mean gradient magnitude	5.66
Maximum gradient	190
Top-hat detected pixels	125
Original intensity range	[20, 220]
Opened intensity range	[20, 200]

6 Discussion

Example 6.1 (Edge Detection via Morphological Gradient) *The morphological gradient* $(f \oplus B) - (f \ominus B)$ *provides a robust edge detector that:*

- *Emphasizes intensity transitions corresponding to object boundaries*
- *Controls edge thickness through structuring element size*
- *Remains less sensitive to noise than derivative-based methods*
- *Produces closed contours around objects*

Remark 6.1 (Structuring Element Selection) *The choice of structuring element shape significantly affects operation results:*

- ***Disk****: Isotropic, no directional bias*
- ***Square****: Emphasizes horizontal/vertical features*
- ***Cross****: Produces connectivity-preserving operations*
- ***Line****: Detects specific orientations*

As demonstrated in Figure 1(p), different SE shapes produce measurable differences in opening results, with disk and square elements differing by 1076 pixels.

Example 6.2 (Top-Hat Transform for Small Feature Detection) *The white top-hat transform* $f-(f \circ B)$ *successfully isolated 125 pixels corresponding to bright features smaller than the structuring element. This operation is particularly valuable for:*

- *Detecting small defects on varying backgrounds*
- *Enhancing low-contrast details*
- *Preprocessing for segmentation*
- *Eliminating uneven illumination effects*

6.1 Algebraic Properties

Theorem 6.1 (Increasing Property) *Both erosion and dilation are increasing operations:*

$$A_1 \subseteq A_2 \Rightarrow (A_1 \ominus B) \subseteq (A_2 \ominus B) \text{ and } (A_1 \oplus B) \subseteq (A_2 \oplus B) \tag{12}$$

Theorem 6.2 (Translation Invariance) *For translation vector* h*:*

$$(A_h \ominus B) = (A \ominus B)_h \quad \text{and} \quad (A_h \oplus B) = (A \oplus B)_h \tag{13}$$

6.2 Application Domains

Morphological operations find extensive use in:

- **Medical imaging**: Vessel segmentation, tumor boundary detection, cell counting
- **Document analysis**: Character recognition, noise removal, skeletonization

- **Industrial inspection**: Defect detection, quality control, measurement
- **Remote sensing**: Road extraction, building detection, land cover classification
- **Video processing**: Object tracking, foreground extraction, motion analysis

7 Conclusions

This analysis demonstrates fundamental morphological operations and their applications:

1. Binary erosion and dilation provide complementary operations for shrinking and expanding objects, with area changes of 40.8% and 47.0% respectively
2. Opening and closing effectively remove noise while preserving essential geometric structure, achieving 98.56% recovery on salt-and-pepper corrupted images
3. Morphological gradient extracts object boundaries with controlled thickness and noise robustness
4. Top-hat transforms isolate small features against varying backgrounds, detecting 125 feature pixels
5. Structuring element shape and size critically influence operation outcomes, requiring careful selection based on application requirements
6. Hit-or-miss transform enables pattern detection for specific geometric configurations

The algebraic properties of morphological operations (duality, idempotence, translation invariance) provide theoretical foundations for developing complex image processing pipelines from fundamental building blocks.

Further Reading

- Soille, P. *Morphological Image Analysis: Principles and Applications*, 2nd ed. Springer, 2003.
- Serra, J. *Image Analysis and Mathematical Morphology.* Academic Press, 1982.

- Gonzalez, R.C. and Woods, R.E. *Digital Image Processing*, 4th ed. Pearson, 2018.
- Haralick, R.M., Sternberg, S.R., and Zhuang, X. "Image analysis using mathematical morphology." *IEEE Trans. Pattern Anal. Mach. Intell.* 9(4): 532-550, 1987.
- Vincent, L. "Morphological grayscale reconstruction in image analysis: applications and efficient algorithms." *IEEE Trans. Image Process.* 2(2): 176-201, 1993.
- Meyer, F. and Beucher, S. "Morphological segmentation." *J. Visual Commun. Image Represent.* 1(1): 21-46, 1990.
- Heijmans, H.J.A.M. *Morphological Image Operators.* Academic Press, 1994.
- Maragos, P. "Pattern spectrum and multiscale shape representation." *IEEE Trans. Pattern Anal. Mach. Intell.* 11(7): 701-716, 1989.
- Najman, L. and Talbot, H. *Mathematical Morphology: From Theory to Applications.* Wiley, 2010.
- Matheron, G. *Random Sets and Integral Geometry.* Wiley, 1975.
- Sternberg, S.R. "Grayscale morphology." *Comput. Vision Graphics Image Process.* 35(3): 333-355, 1986.
- Salembier, P. and Serra, J. "Flat zones filtering, connected operators, and filters by reconstruction." *IEEE Trans. Image Process.* 4(8): 1153-1160, 1995.
- Roerdink, J.B.T.M. and Meijster, A. "The watershed transform: definitions, algorithms and parallelization strategies." *Fundam. Inform.* 41(1-2): 187-228, 2000.
- Beucher, S. and Meyer, F. "The morphological approach to segmentation: the watershed transformation." In *Mathematical Morphology in Image Processing*, CRC Press, 1993.
- Adams, R. and Bischof, L. "Seeded region growing." *IEEE Trans. Pattern Anal. Mach. Intell.* 16(6): 641-647, 1994.
- Boomgaard, R. and Smeulders, A. "The morphological structure of images: the differential equations of morphological scale-space." *IEEE Trans. Pattern Anal. Mach. Intell.* 16(11): 1101-1113, 1994.
- Jackway, P.T. and Deriche, M. "Scale-space properties of the multiscale morphological dilation-erosion." *IEEE Trans. Pattern Anal. Mach. Intell.* 18(1): 38-51, 1996.

- Cheng, F. and Venetsanopoulos, A.N. "An adaptive morphological filter for image processing." *IEEE Trans. Image Process.* 1(4): 533-539, 1992.
- Cuisenaire, O. and Macq, B. "Fast Euclidean distance transformation by propagation using multiple neighborhoods." *Comput. Vision Image Understand.* 76(2): 163-172, 1999.
- Dougherty, E.R. and Lotufo, R.A. *Hands-on Morphological Image Processing.* SPIE Press, 2003.